Product Requirements Document (PRD)

Project Name: Fintech360 Data Pipeline **Document Owner:** [Your Name/Title] **Status:** Draft / Under Review **Date:** October 2026

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1. Executive Summary & Problem Statement

Currently, our business teams (Risk, Underwriting, and Collections) are operating on stale data. The existing reporting infrastructure queries our primary transactional database (AWS RDS) directly, which causes performance bottlenecks during peak hours and delays report generation by hours.

If a loan account slips into early delinquency (DPD), our collections team often doesn't see it until the next day. Furthermore, bad data (nulls, duplicates) is occasionally making its way into downstream risk models.

Fintech360 is a scalable, Medallion-architecture data pipeline built on Databricks and AWS. It physically separates transactional processing from analytical workloads. By implementing strict data quality gates and a quarantine system, we will ensure that only 100% validated, near real-time data reaches our Power BI dashboards and risk analysts.

2. Target Audience

- **Credit Risk Team:** Needs clean, historical datasets to train risk models.
- **Underwriting Team:** Needs rapid lookups for customer credit profiles.
- **Collections Team:** Requires up-to-date daily dashboards for roll-rate tracking and agent allocation.
- **Data Science / Analysts:** Needs a governed environment (Unity Catalog) to run ad-hoc queries without bringing down production databases.

3. Objectives & Key Results (OKRs)

- **Objective 1: Eliminate data latency for critical business operations.**
    - *KR1:* Reduce dashboard refresh time from hours to under 15 minutes.
    - *KR2:* Achieve 99.9% uptime for the daily ingestion pipeline.
- **Objective 2: Guarantee data integrity for financial reporting.**
    - *KR1:* Implement 100% automated schema and data quality validation before data hits the Silver layer.
    - *KR2:* Reduce data-related BI incidents (e.g., "the numbers look wrong") by 90% in the first quarter.

- **Objective 3: Ensure strict data governance.**
    - *KR1:* Mask 100% of PII (Personally Identifiable Information) in the reporting (Gold) layer using Unity Catalog.

4. Scope

In Scope

- Extraction of data from AWS RDS to S3 Landing Zone.
- Databricks Medallion pipeline (Bronze, Silver, Gold).
- Automated Data Quality Gates (Null checks, Duplicate checks, Referential integrity).
- Quarantine/Dead-letter routing for failed records.
- Unity Catalog implementation for Row/Column-level security.
- Power BI semantic model optimization connected to the Gold layer.

Out of Scope

- Modifying the upstream Loan Origination System (LOS) architecture.
- Developing new Machine Learning models (we are only providing the feature store/data foundation).

5. Functional Requirements

- **F1 - Change Data Capture (CDC):** The pipeline must capture ongoing inserts, updates, and deletes from the source RDS without requiring full table scans every day.
- **F2 - Schema Evolution Handling:** The Bronze-to-Silver transition must automatically detect if a new column is added upstream. It should validate data types before proceeding.
- **F3 - Data Quarantine:** Any record failing the defined business rules (e.g., negative loan amount, missing customer ID) must be routed to a Quarantine table. It must *not* fail the entire batch run.
- **F4 - Error Monitoring:** The system must trigger an alert (via Slack/Email) to the Data Engineering team whenever the Quarantine table receives new records above a 1% threshold of the batch size.
- **F5 - Business Transformations:** The Gold layer must pre-calculate complex metrics like Exposure at Default (EAD), Collection Efficiency, and 30+ DPD flags so Power BI doesn't have to compute them on the fly.

6. Non-Functional Requirements

- **Scalability:** The pipeline must auto-scale compute clusters during month-end processing when transaction volumes spike by up to 300%.

- **Security:** All data at rest in S3 must be encrypted. Unity Catalog must enforce Role-Based Access Control (RBAC).
- **Performance:** End-user queries in Power BI hitting the Databricks SQL Warehouse must return results in under 3 seconds.

7. Assumptions & Dependencies

- We have read-replica or CDC access (e.g., AWS DMS) to the RDS source to prevent impacting production performance during extraction.
- The business intelligence team has Power BI Premium or Pro licenses to utilize DirectQuery/DirectLake features.
- Downstream teams agree to standard data definitions (e.g., everyone agrees on how "Collection Efficiency" is calculated) before we build the Gold layer.